B.Tech 3rd Sem: Discrete Mathematics
Detailed Subjective Solutions

BEU PATNA

Note: All solutions are presented step-by-step. Each sub-question starts on a new page.

Question 2 (a)

Problem: Define an Equivalence Relation. Let R be a relation on the set of integers $\mathbb{Z}$ defined by xRy if and only if $(x-y)$ is divisible by 5. Prove that R is an equivalence relation.

Step-by-Step Solution:

Definition: A relation R on a set A is an **Equivalence Relation** if it satisfies three properties: 1. **Reflexivity:** aRa for all $a \in A$. 2. **Symmetry:** If aRb, then bRa. 3. **Transitivity:** If aRb and bRc, then aRc.

Proof for the given relation: Let $x, y, z \in \mathbb{Z}$. The relation is $xRy \iff 5 \mid (x-y)$.

1. **Reflexive Property:** We check if xRx is true. Calculation: $x - x = 0$. Since 0 is divisible by 5 (i.e., $0 = 5 \times 0$), $5 \mid (x-x)$. Hence, R is reflexive.
2. **Symmetric Property:** Assume xRy. This means $5 \mid (x-y)$. So, $x - y = 5k$ for some integer k. Multiply both sides by -1: $-(x-y) = -5k \implies y-x = 5(-k)$. Since $-k$ is also an integer, $5 \mid (y-x)$. This means yRx. Hence, R is symmetric.
3. **Transitive Property:** Assume xRy and yRz. This means $5 \mid (x-y)$ and $5 \mid (y-z)$. So, $x-y = 5k$ and $y-z = 5m$ for integers k, m. Adding these equations: $(x-y)+(y-z) = 5k+5m$ $x-z = 5(k+m)$. Since $(k+m)$ is an integer, $5 \mid (x-z)$. This means xRz. Hence, R is transitive.

Conclusion: Since the relation R is reflexive, symmetric, and transitive, it is an equivalence relation.

Question 2 (b)

Problem: Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \frac{3x-4}{5}$. Show that f is a bijection and find its inverse f^{-1}.

Step-by-Step Solution:

A function is a **bijection** if it is both Injective (One-to-One) and Surjective (Onto).

Step 1: Prove Injective (One-to-One) Let $f(x_1) = f(x_2)$ for $x_1, x_2 \in \mathbb{R}$.

$$\begin{aligned} \frac{3x_1 - 4}{5} &= \frac{3x_2 - 4}{5} \\ 3x_1 - 4 &= 3x_2 - 4 \quad \text{(Multiplying by 5)} \\ 3x_1 &= 3x_2 \quad \text{(Adding 4)} \\ x_1 &= x_2 \quad \text{(Dividing by 3)} \end{aligned}$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$, the function is injective.

Step 2: Prove Surjective (Onto) Let y be an arbitrary element in the codomain $\mathbb{R}$. We need to find $x \in \mathbb{R}$ such that $f(x) = y$.

$$\begin{aligned} y &= \frac{3x - 4}{5} \\ 5y &= 3x - 4 \\ 3x &= 5y + 4 \\ x &= \frac{5y + 4}{3} \end{aligned}$$

Since for every $y \in \mathbb{R}$, there exists a real number $x = \frac{5y+4}{3}$ such that $f(x) = y$, the function is surjective.

Conclusion on Bijection: Since f is both injective and surjective, it is a bijective function.

Step 3: Find the Inverse f^{-1} To find the inverse, we use the expression for x found in Step 2:

$$f^{-1}(y) = \frac{5y + 4}{3}$$

Replacing y with x as the standard variable:

$$f^{-1}(x) = \frac{5x + 4}{3}$$

Question 3 (a)

Problem: Find the GCD of 81 and 237 using the Euclidean Algorithm and express it in the form $81x + 237y$, where x and y are integers.

Step-by-Step Solution:

Part 1: Finding GCD using Euclidean Algorithm We divide the larger number by the smaller one: 1. $237 = 2 \times 81 + 75$ (Equation A) 2. $81 = 1 \times 75 + 6$ (Equation B) 3. $75 = 12 \times 6 + 3$ (Equation C) 4. $6 = 2 \times 3 + 0$ (Stop)

The last non-zero remainder is **3**. So, GCD$(81, 237) = 3$.

Part 2: Expressing as $81x + 237y$ (Back-substitution) We use the equations from above, starting from the one with remainder 3: From (C): $3 = 75 - 12 \times 6$

Substitute 6 from (B) ($6 = 81 - 75$): $3 = 75 - 12 \times (81 - 75)$ $3 = 75 - 12 \times 81 + 12 \times 75$ $3 = 13 \times 75 - 12 \times 81$

Substitute 75 from (A) ($75 = 237 - 2 \times 81$): $3 = 13 \times (237 - 2 \times 81) - 12 \times 81$ $3 = 13 \times 237 - 26 \times 81 - 12 \times 81$ $3 = 13 \times 237 - 38 \times 81$

Final Form: $3 = 81(-38) + 237(13)$ Thus, $x = -38$ and $y = 13$.

Question 3 (b)

Problem: State and prove the Fundamental Theorem of Arithmetic.

Step-by-Step Solution:

Statement: Every integer $n > 1$ is either a prime number itself or can be represented as the product of prime numbers, and this representation is unique, except for the order in which the prime factors are written.

Proof: The proof consists of two parts: Existence and Uniqueness.

1. Existence (by Strong Induction):

- **Base Case:** Let $n = 2$. 2 is prime, so it is a product of primes (just one).
- **Inductive Step:** Assume every integer k such that $2 \leq k < n$ can be factored into primes.
- **Case 1:** If n is prime, we are done.
- **Case 2:** If n is composite, $n = a \times b$ where $1 < a, b < n$. By our assumption, both a and b can be written as products of primes. Therefore, $n = (\text{product of primes}) \times (\text{product of primes})$ is also a product of primes.

2. Uniqueness (using Euclid's Lemma): Euclid's Lemma states: If a prime p divides ab, then $p \mid a$ or $p \mid b$. Assume n has two different factorizations: $n = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$

- Since p_1 divides n, p_1 must divide the product $q_1 q_2 \dots q_s$.
- By Euclid's Lemma, p_1 must divide at least one q_j.
- Since both p_1 and q_j are prime, $p_1 = q_j$.
- We cancel p_1 and q_j from both sides and repeat the process.
- Eventually, all factors will be paired up, showing that the two factorizations were identical (except for order) and $r = s$.

Question 4 (a)

Problem: Using the Principle of Mathematical Induction, prove that for all $n \geq 1$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Step-by-Step Solution:

Let $P(n)$ be the statement: $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Step 1: Base Case ($n = 1$) LHS = $1^2 = 1$ RHS = $\frac{1(1+1)(2(1)+1)}{6} = \frac{1\times2\times3}{6} = \frac{6}{6} = 1$ LHS = RHS. So, $P(1)$ is true.

Step 2: Inductive Hypothesis Assume $P(k)$ is true for some integer $k \geq 1$. That is, $1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$

Step 3: Inductive Step We must prove that $P(k+1)$ is true. LHS for $P(k+1) = (1^2 + 2^2 + \cdots + k^2) + (k+1)^2$ Using the assumption from Step 2: LHS = $\frac{k(k+1)(2k+1)}{6} + (k+1)^2$ Take $(k+1)$ common: $= (k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right] = (k+1)\left[\frac{2k^2+k+6k+6}{6}\right] = \frac{(k+1)(2k^2+7k+6)}{6}$ Factoring the quadratic $2k^2+7k+6$: $2k^2+4k+3k+6 = 2k(k+2)+3(k+2) = (k+2)(2k+3)$ So, LHS = $\frac{(k+1)(k+2)(2k+3)}{6}$ This is exactly the RHS of $P(k+1)$ where n is replaced by $k+1$.

Conclusion: Since $P(1)$ is true and $P(k) \implies P(k+1)$, by PMI, the statement is true for all $n \geq 1$.

Question 4 (b)

Problem: In a survey of 100 students, it was found that 50 play Football, 45 play Cricket, and 40 play Hockey. 20 play both Football and Cricket, 15 play Cricket and Hockey, and 10 play Football and Hockey. 5 students play all three games. Find the number of students who do not play any of the three games.

Step-by-Step Solution:

Let: F: Set of students playing Football ($|F| = 50$) C: Set of students playing Cricket ($|C| = 45$) H: Set of students playing Hockey ($|H| = 40$)

Intersections given: $|F \cap C| = 20$ $|C \cap H| = 15$ $|F \cap H| = 10$ $|F \cap C \cap H| = 5$

Step 1: Calculate the number of students playing at least one game. We use the Principle of Inclusion-Exclusion for three sets:

$$|F \cup C \cup H| = |F| + |C| + |H| - (|F \cap C| + |C \cap H| + |F \cap H|) + |F \cap C \cap H|$$

Substitute the values: $|F \cup C \cup H| = 50 + 45 + 40 - (20 + 15 + 10) + 5$ $|F \cup C \cup H| = 135 - 45 + 5$ $|F \cup C \cup H| = 90 + 5 = 95$

Step 2: Calculate students who do not play any game. Total students (U) = 100. Students playing none = $|U| - |F \cup C \cup H|$ Students playing none = $100 - 95 = 5$

Final Answer: 5 students do not play any of the three games.

Question 5 (a)

Problem: Construct a truth table to verify the logical equivalence:

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$$

Step-by-Step Solution:

We will construct the truth table column by column.

p	q	$\neg p$	$\neg q$	$(\neg p \wedge q)$	$p \vee (\neg p \wedge q)$	**LHS:** $\neg(p \vee (\neg p \wedge q))$	**RHS:** $\neg p \wedge \neg q$
T	T	F	F	F	T	**F**	**F**
T	F	F	T	F	T	**F**	**F**
F	T	T	F	T	T	**F**	**F**
F	F	T	T	F	F	**T**	**T**

Observations: 1. Column 7 represents the Truth Values for the LHS: $\neg(p \vee (\neg p \wedge q))$. 2. Column 8 represents the Truth Values for the RHS: $\neg p \wedge \neg q$.

Conclusion: Since the truth values in Column 7 and Column 8 are identical for all possible combinations of p and q, the logical equivalence is verified.

Question 5 (b)

Problem: Explain the Rules of Inference. Use them to show that the following premises lead to the conclusion r: $p \lor q$, $q \to r$, and $\neg p$.

Step-by-Step Solution:

Rules of Inference are logical templates used to derive a conclusion from a set of premises. Common rules include: 1. **Modus Ponens:** If p and $p \to q$ are true, then q is true. 2. **Disjunctive Syllogism:** If $p \lor q$ is true and $\neg p$ is true, then q is true. 3. **Hypothetical Syllogism:** If $p \to q$ and $q \to r$ are true, then $p \to r$ is true.

Applying rules to the given problem:
Premises: 1. $p \lor q$ 2. $q \to r$ 3. $\neg p$
Step-by-step derivation:

1. **Premise 1:** $p \lor q$
2. **Premise 3:** $\neg p$
3. **Result 1:** From (1) and (2), using **Disjunctive Syllogism**, we get: q
4. **Premise 2:** $q \to r$
5. **Result 2:** From Result 1 (q) and Premise 2 ($q \to r$), using **Modus Ponens**, we get: r

Conclusion: The given premises logically lead to the conclusion r.

Question 6 (a)

Problem: Let $(G, \cdot)$ be a group. Prove that for any $a, b \in G$:

$$(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$$

Step-by-Step Solution:

In a group, the inverse of an element x is unique and satisfies $x \cdot x^{-1} = e$, where e is the identity element. To prove $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$, we must show that multiplying $(a \cdot b)$ by $(b^{-1} \cdot a^{-1})$ results in the identity element e.

Case 1: Right-hand multiplication

$$\begin{aligned}(a \cdot b) \cdot (b^{-1} \cdot a^{-1}) &= a \cdot (b \cdot b^{-1}) \cdot a^{-1} \quad \text{(Associative property)} \\ &= a \cdot e \cdot a^{-1} \quad \text{(Property of inverse: } b \cdot b^{-1} = e\text{)} \\ &= a \cdot a^{-1} \quad \text{(Property of identity)} \\ &= e \quad \text{(Property of inverse)}\end{aligned}$$

Case 2: Left-hand multiplication

$$\begin{aligned}(b^{-1} \cdot a^{-1}) \cdot (a \cdot b) &= b^{-1} \cdot (a^{-1} \cdot a) \cdot b \quad \text{(Associative property)} \\ &= b^{-1} \cdot e \cdot b \\ &= b^{-1} \cdot b \\ &= e\end{aligned}$$

Conclusion: Since $(a \cdot b) \cdot (b^{-1} \cdot a^{-1}) = e$ and $(b^{-1} \cdot a^{-1}) \cdot (a \cdot b) = e$, it follows that $b^{-1} \cdot a^{-1}$ is the inverse of $a \cdot b$. Thus, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.

Question 6 (b)

Problem: Define a Subgroup. Prove that a non-empty subset H of a group G is a subgroup if and only if for all $a, b \in H, ab^{-1} \in H$.

Step-by-Step Solution:

Definition: A non-empty subset H of a group G is called a **subgroup** if H itself forms a group under the same operation as G.

Proof of the One-Step Subgroup Test: The statement is "if and only if", so we prove both directions.

Direction 1 ($\implies$): Assume H is a subgroup. Since H is a subgroup: 1. For any $b \in H$, its inverse b^{-1} must also be in H. 2. Since H is closed under the group operation, for any $a \in H$ and $b^{-1} \in H$, the product ab^{-1} must be in H. So, $ab^{-1} \in H$ holds.

Direction 2 ($\impliedby$): Assume for all $a, b \in H$, $ab^{-1} \in H$. We need to prove H satisfies group axioms:

1. **Identity:** Since H is non-empty, let $x \in H$. Applying the condition ab^{-1} with $a = x$ and $b = x$: $xx^{-1} = e \in H$. So identity is present.

2. **Inverse:** For any $x \in H$, we know $e \in H$. Applying the condition with $a = e$ and $b = x$: $ex^{-1} = x^{-1} \in H$. So inverse exists for every element.

3. **Closure:** Let $x, y \in H$. We know $y^{-1} \in H$. Applying the condition with $a = x$ and $b = y^{-1}$: $x(y^{-1})^{-1} = xy \in H$. So H is closed.

4. **Associativity:** It is inherited from G.

Conclusion: H is a subgroup of G.

Question 7 (a)

Problem: Simplify the following Boolean expression using Boolean Algebra laws:

$$F(x, y, z) = x'yz + x'yz' + xy'z' + xy'z$$

Step-by-Step Solution:

Given Expression: $F = x'yz + x'yz' + xy'z' + xy'z$

Step 1: Grouping terms We can group the first two terms and the last two terms: $F = (x'yz + x'yz') + (xy'z' + xy'z)$

Step 2: Apply Distributive Law Factor out $x'y$ from the first group and xy' from the second group: $F = x'y(z + z') + xy'(z' + z)$

Step 3: Apply Complement Law We know that in Boolean Algebra, $z + z' = 1$: $F = x'y(1) + xy'(1)$

Step 4: Apply Identity Law Anything ANDed with 1 remains itself: $F = x'y + xy'$

Step 5: Identify the gate (Optional) The expression $x'y + xy'$ represents the **XOR** (Exclusive OR) operation between x and y.

Final Simplified Expression:

$$F = x'y + xy'$$

Question 8 (a)

Problem: State and prove Euler's Formula for a connected planar graph with V vertices, E edges, and F faces.

Step-by-Step Solution:

Statement: For any connected planar graph:

$$V - E + F = 2$$

Proof (by Induction on the number of Edges E):

Base Case ($E = 0$): For a connected graph with 0 edges, there must be only 1 vertex ($V = 1$). The graph has only 1 face (the outer infinite face), so $F = 1$. Checking the formula: $V - E + F = 1 - 0 + 1 = 2$. True for $E = 0$.

Inductive Hypothesis: Assume the formula $V - E + F = 2$ holds for any connected planar graph with k edges.

Inductive Step: Consider a connected planar graph G with $k + 1$ edges.

- **Case 1: G is a tree.** In a tree, $E = V - 1$ and $F = 1$. $V - (V - 1) + 1 = V - V + 1 + 1 = 2$. Formula holds.
- **Case 2: G contains a cycle.** Let e be an edge in a cycle. e separates two faces. If we remove edge e to get G', then: $V' = V$ (vertices remain same) $E' = E - 1$ (one edge removed) $F' = F - 1$ (the two faces separated by e merge into one) Since G' has k edges, the hypothesis applies: $V' - E' + F' = 2$. Substitute the values: $V - (E - 1) + (F - 1) = 2$ $V - E + 1 + F - 1 = 2$ $V - E + F = 2$.

Conclusion: Euler's formula is proved.

Question 8 (b)

Problem: Define Eulerian and Hamiltonian graphs. Provide an example of a graph that is Eulerian but not Hamiltonian.

Step-by-Step Solution:

1. Eulerian Graph: A connected graph is Eulerian if it contains a closed trail (Eulerian Circuit) that includes **every edge** of the graph exactly once. *Condition:* A connected graph is Eulerian if and only if every vertex has an **even degree**.

2. Hamiltonian Graph: A graph is Hamiltonian if it contains a cycle (Hamiltonian Cycle) that visits **every vertex** exactly once and returns to the starting vertex. *Condition:* There is no simple characterization like Eulerian graphs, though Ore's and Dirac's theorems provide sufficient conditions.

3. Example: Eulerian but NOT Hamiltonian

Consider the "Butterfly Graph" or "Hourglass Graph" (two triangles sharing a common vertex).

- **Structure:** Let vertices be $\{A, B, C, D, E\}$. Let C be the center vertex. Edges: $(A, B), (B, C), (C, A)$ (Triangle 1) and $(C, D), (D, E), (E, C)$ (Triangle 2).
- **Degrees:** $\deg(A) = 2, \deg(B) = 2, \deg(C) = 4, \deg(D) = 2, \deg(E) = 2.$
- **Why Eulerian?** All vertices have even degrees. We can start at C, traverse Triangle 1, return to C, traverse Triangle 2, and end at C.
- **Why NOT Hamiltonian?** To visit all vertices, we must visit C to move from Triangle 1 to Triangle 2. But to complete a cycle, we would have to pass through C a second time before returning to the start, which violates the Hamiltonian cycle definition (visiting each vertex exactly once).

Question 9 (a)

Problem: Prove by contradiction that $\sqrt{2}$ is an irrational number.

Step-by-Step Solution:

Assumption: Assume $\sqrt{2}$ is a **rational** number.

1. By definition of rational numbers, we can write: $\sqrt{2} = \frac{p}{q}$ where p and q are integers, $q \neq 0$, and they are in **simplest form** (co-prime, meaning they have no common factors other than 1).
2. Square both sides: $2 = \frac{p^2}{q^2} \implies p^2 = 2q^2$.
3. This means p^2 is an **even** number. Since the square of an odd number is odd, p itself must be **even**.
4. Let $p = 2k$ for some integer k. Substitute this in the equation $p^2 = 2q^2$: $(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2$.
5. This means q^2 is also an **even** number. Consequently, q must be **even**.
6. Now we have found that both p and q are even, which means they both have **2** as a common factor.
7. This **contradicts** our original assumption that p and q are co-prime (have no common factors).

Conclusion: Since the assumption leads to a contradiction, it must be false. Therefore, $\sqrt{2}$ is an irrational number.

Question 9 (b)

Problem: What is a Rooted Tree? Define the following terms with respect to trees: (i) Level of a node (ii) Height of a tree (iii) Leaf node (iv) Internal node.

Step-by-Step Solution:

Rooted Tree: A rooted tree is a tree in which one vertex has been designated as the **root**, and every edge is directed away from the root. It creates a hierarchy of parent-child relationships.

Definitions:

(i) **Level of a node:** The level of a node is defined as the length of the path from the root to that specific node.

- The root is at **Level 0**.
- Children of the root are at Level 1, their children at Level 2, and so on.

(ii) **Height of a tree:** The height of a tree is the **maximum level** of any node in that tree. Alternatively, it is the number of edges on the longest path from the root to a leaf.

(iii) **Leaf node:** A leaf node (also called an external node or terminal node) is a node that has **no children**. In terms of degrees, it is a vertex with degree 1 (unless it is the root of a single-node tree).

(iv) **Internal node:** An internal node is a node that has **at least one child**. In a rooted tree, all nodes except the leaves are internal nodes. The root is an internal node if the tree has more than one node.